



USDT0 Audit Report

TONTECH VERIFIED ON 11 JULY 2025

01 Summary

Total findings	Resolved	Acknowledged	Declined	Unresolved
1	0	0	0	1

The findings were classified by the severity levels as following:

Critical	Major	Medium	Minor	Informational
0	0	0	0	1

02 Severity levels

Critical

These are the most severe security flaws in a smart contract. They can lead to immediate and significant risks, allowing attackers to gain unauthorized access, bypass security mechanisms, or cause substantial financial losses. Critical vulnerabilities require immediate attention and remediation.

Major

These vulnerabilities are significant but may not pose an immediate threat or have a lower probability of exploitation compared to critical vulnerabilities. However, they still have the potential to result in significant damages or security breaches if left unaddressed. They should be resolved in a timely manner to ensure the overall security of the smart contract.

Medium

Medium vulnerabilities indicate security weaknesses that may not have a high likelihood of exploitation or immediate serious consequences. However, they can still impact the overall security posture of the smart contract and should be fixed to enhance its resilience against potential attacks or vulnerabilities.

Minor

These vulnerabilities are less critical in nature and generally have a limited impact on the security of the smart contract. While they may not pose an immediate threat, they should not be overlooked, as they can compound with other vulnerabilities or serve as a stepping stone for potential attackers. Addressing them is important to maintain a robust security posture.

Informational

Suggestions for code style, architecture, and optimization to improve quality and efficiency. Not security risks, but enhance resilience.

03 Codebase

[Repository](#) ([LayerZero-Labs/x-usdtoft](#))

[Commit \(EVM\)](#) ([ad75806292bfb1596e2da77a6b9026f9e6cb0c3f](#))

[Commit \(TON\)](#) ([ecc5fc8eac585b23f78bc08599ce2a106980ba15](#))

04 Scope

EVM

File path	SHA-256 hash of the file
contracts/IMultiHopComposer.sol	4e640413c661ef853a8b68f431921a3532f791b23716d9304dd1c0e54c53d895
contracts/IUsdtOFT.sol	b0d500ac1be0f0fb103dba83c69b73e260ae4c378431bcd680f82f7a0721463

File path	SHA-256 hash of the file
contracts/MultiHopComposer.sol	6c65327de9f4033c0366a21f5fb63c57612e0345285727140dcc3ac9f8cd90ce
contracts/UsdtOFT.sol	a7ebdb0590f43d74cc647a376cb3819d6b33b1c360f4e4f64b2a92ba189b5a5f
test/foundry/mock/ERC20Mock.sol	443ede2b21b8c72482f5ed90ead75a6f8dd7a2b01d89104e89efe1e8d36eb58e
test/foundry/mock/OFTMock.sol	1a3c05bb4622d8092dbfb970e0d9375fb9247d55b59c5f650e10824f066668f1
test/foundry/MultiHopComposer/MultiHopComposer_E2E.t.sol	525e2340ab3b6926191793178c22a0e4d8b0c3a5b85f63681a29990074f11c50
test/foundry/MultiHopComposer/MultiHopComposer_Unit.t.sol	d28ee71c2b12c13348149c7a272e22832abcdaca7e2f1f009fa2113b038c8531
test/foundry/MultiHopComposer/MultiHopComposerBase.t.sol	8893283f5deb0d21b1b8f209fedc160b895a233e07228065552dadac68616555
test/foundry/UsdtOFT.t.sol	47e1c9fd0fb099661d390b063a116c7a47f74b022fe98fbafd2d95381b5c1ae4

TON

File path	SHA-256 hash of the file
src/handler.fc	03264d58182b413d5f99bcb8b3b0ea8e5f5975214326aa1ca2fdaf228ff17e03
src/main.fc	621c55d132cff58db6dc6616cf6d43671d4c57dc7251265395d391b7cbb7b5f0
src/oApp/handlerOApp.fc	98a86de7373719a469529744c2ca653090bb8278f3724c3d4f243a9a24747987

File path	SHA-256 hash of the file
src/oApp/interface.fc	b2644065ffbb16fb9238adae9daf655a e93335f7bd3fa6b0e1b7e203ec204050
src/oApp/storage.fc	c184a7048cb0e851ba4fdfcc67b05f52 1d8fd21cefa63a9272aa5cd730f9f4a6
src/usdt0FT/coders.fc	66cc138836d2920b32a139f544e2971f 31ee1ec7437b77679ff589c44cfcb5c9
src/usdt0FT/consts.fc	cb2086a7c70a3239adfe8a3d68be1d21 d35d2e7b784dc93779cd617743c24761
src/usdt0FT/handlerUsdt.fc	c30710eeb76b244641ec5f388d13b940 b29a3e0ec812ebed98f21d615be6a670
src/usdt0FT/interface.fc	e364ce55b6ec201aca492054f53bcd1 38e10f7511789a43ef73349baf945f49
src/usdt0FT/storage.fc	67353000b2a0e971d7d70fe99c7e7acf c12ba7c30e07665269fca168c6cbc23c
structs/Address.fc	1a22f0a2583d8646cb49192f939f6396 2869d6352785bd043d0b914bb5e46dec
structs/Amount.fc	8bdeb74245a1a170b7a16fe4e2d600c2 c4aeca75cee494097e70e397d0be74b8
structs/CostAsserts.fc	e9306bc0b391e3d4c10c1d774f748d2d 3c830372d2486f84364fd4fc46c1acc8
structs/Credits.fc	7bd168d271e3cc5b59d59498c78eee52 49e0eb2d01393582c0f80fd74141df45
structs/Fee.fc	9a7430fa9e8f6cd071fdbb86a478f13a 2ebb1902497e796f7b504967bcfb6cd8
structs/GasAsserts.fc	361b40638d5026a28719b44961549926 abc466c7458a4b0c25a1fbf3a0da5bec

File path	SHA-256 hash of the file
structs/LayerZeroStructs.fc	0b91b830f0e1fc1b04f9dd78bd178b3c c5523d1adcb684be09ed1b84d5a4f7ef
structs/MdError.fc	86dfa9defd0a581dc6d9a0d8fd722a90 120511df172f67d63150fb8cae02e798
structs/MdGuid.fc	3672317dded65b2cf51fe8f9c913d3c1 2223e7172a21ac7b3358ae2dc0ee181c
structs/MsgSize.fc	789c4fa3756a1e050cdc9326b4843c8 d961508681ecf52e2897570f52e2b210
structs/OFTCredits.fc	83bce25ccad058850d5391499274bb22 9917cb4a61a1e4c3dbf33de71777c83e
structs/OFTSend.fc	5270c8d76cfde74726d8c51a77b725ed ed71018c2065a6c6e147be9fbdbcf80d
structs/RecoverUsdt.fc	75b7528058753578a96c1b4ad97045f0 71c76f90aeb649b7779af7641bb9e8b7
structs/SetPeer.fc	7cf84425db2f7e41a4ea0fd977c56096 e7a3ab573801680843b23b67ab73a577
structs/TokenTransfer.fc	8f710c0b2dbbb56514a440fa9d07bb1a 0ffae27f89f82a1b6337157be60c3094
tests/baseSerdeTest.fc	949b329fffb32da026339b5ac4f4be8dd 5328295b1f8a84719fc745b1f2a42cea
tests/consts.fc	6297dc61041c5fd4aef297b9e0fbb492 d3c665fe80f1e74d3dc2449b89879a54
tests/mocks.fc	981146bf582b180342a7035b9a7ab7e3 a89dd6f7fd871249440b9ef0ce61f4ca
tests/testUtils.fc	3322f95d98c7a7a638e3499066f693ee d071dcd92c377767cc7309465ad4b9f5

File path	SHA-256 hash of the file
tests/usdtOFT/channel.fc	b485883e7b439293e715274b8b915104 2299c4c45cd3d49cc73d17440dbf562b
tests/usdtOFT/credits.fc	f3e4f67eb352bc30b43fa3a2864e10cc 6b68916863eda4ac22f9f2c849462459
tests/usdtOFT/helpersCredits.fc	8f8ed0374698355f7d745266ec2854ef 24c20aa06070dcb1a9c97c0f50cc3458
tests/usdtOFT/helpersOApp.fc	2c4de8b23255e4be9bde30b32151decc 50e1e65b1e488176e3893444e03140f0
tests/usdtOFT/helpersOFT.fc	26963f708bd81fa98a1ff6ad2b941522 7e2510b04a5b1df1806f18a7da84e1e2
tests/usdtOFT/liquidity.fc	cbb237deeb759d659cab446ab3398f74 f98664153e1a03f1efceca239f369fd2
tests/usdtOFT/oft.fc	cdb20f1bc01d1efa7cc16d14d707ec64 32eb1f11cb78ef67b0c0155ec5e805ab
tests/usdtOFT/optionsOApp.fc	50ec6bbbeb5df0c354cb5d29df9a2bfd9 e834f7609a0a36344da3e585f9ef52cf
tests/usdtOFT/ownershipOApp.fc	05904deb8e0e5a130e5c54f938175590 fc504dda19df6265403400854dad10c2
tests/usdtOFT/ownershipOFT.fc	de7cb7a1075aa2990b0f23d19b9b9cb5 1f24314cb417d3f6d601ace416bedc2e
tests/usdtOFT/permissions.fc	6139678b7f0b189e430b1b4464e588eb f70a82f40e4209e69a82560a6b7ea485
tests/usdtOFT/recoverUsdt.fc	73ef34719a1ec7c7ca66e89f74000a4a b30ab3110079b3b62e41922414fac6a2
tests/usdtOFT/serde/ addressSerde.fc	99486534ce255262a3477a1c089de5ed 319bc9cca34e97617cf8ae8bad41741d

File path	SHA-256 hash of the file
tests/usdtOFT/serde/ amountSerde.fc	62c63f72e47529b308b301229d99379b e8079527a3d74b73ecee240e4a2850f5
tests/usdtOFT/serde/ costAssertsSerde.fc	091b01d232b70b2981ab1ed642cd5d07 d2b19e4f55c3186eb6aff3178b266c91
tests/usdtOFT/serde/ creditsSerde.fc	f4353dd837fc8fe67591c044705e6d7a b15699f8ef3d342ba99c7ae639553163
tests/usdtOFT/serde/ feeSerde.fc	e33bd501f9ead0e631d737334a936d7c a5eaf59fe2fe2ec3d62b85ee9f6bc4f2
tests/usdtOFT/serde/ gasAssertsSerde.fc	86a6849ddb6b12042f74ae97a6f7803c 545913c0299770d2d60a29235b12424a
tests/usdtOFT/serde/ mdErrorSerde.fc	ecf3b194afcead460efe11e9dbade4ef e35a86a7b097481d4741b12317ed9d0f
tests/usdtOFT/serde/ mdGuidSerde.fc	7f771e1f504855cd17a0aa443f772c78 e3e40ae01665fb15d364669e69e30038
tests/usdtOFT/serde/ oAppStorageSerde.fc	ab35cac8a0f4603d0d97898f128e6056 ed278f3431e031c093c2411a3b381c97
tests/usdtOFT/serde/ oftCreditsSerde.fc	d06d735085d7fa53599e3bd210d56ab6 9373772e1b8e8d7b2332d66bd7b75d9a
tests/usdtOFT/serde/ oftSendSerde.fc	8b698557ad3f0a9248b6a7a74dfec23c 3bedfd0568237355ed03b3713e9b6f41
tests/usdtOFT/serde/ recoverUsdtSerde.fc	a4d3b7b78c82fe48494ac3ebcd732fe6 21b44257aa83619879e5b295d26668a6
tests/usdtOFT/serde/ setPeerSerde.fc	dd7f3c0f0fb5b7a6334b4ff014bec515 6ff21ad989c6b1468475cce753a46f0d
tests/usdtOFT/serde/ tokenTransferSerde.fc	100ee9d18560a7c0c3413c55b709e546 7fffee13792a9a37aa27ef93ba3fce58

File path	SHA-256 hash of the file
tests/usdt0FT/serde/ usdt0ftStorageSerde.fc	89bd494c2f3685dcfe16228fd9541110 f2a004aacb7625fe3f92b1137f3057c2

05 Issues

Unintentionally high price ratio can lead to error confusion

Category	Severity	Location	Status
Logic error	Informational	src/usdtOFT/handlerUsdt.fc L342-388	Unresolved

Description

In `_assertOptionsCost`, `maxPriceRatio` is initially set to `MAX_U256`, and later set to the actual value depending on the destination endpoint ID. It is then used in a multiplication, and so if the value is still `MAX_U256`, it is inferred that the endpoint ID is unknown. However, if the price ratio is too high at any point, the same error will be thrown.

Recommendation

Throw a specific error when the destination endpoint ID is unknown and use the `muldiv` operation for the multiplication and division.